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Nontarget Screening of Per- and Polyfluoroalkyl Substances Binding to Human Liver Fatty Acid Binding Protein

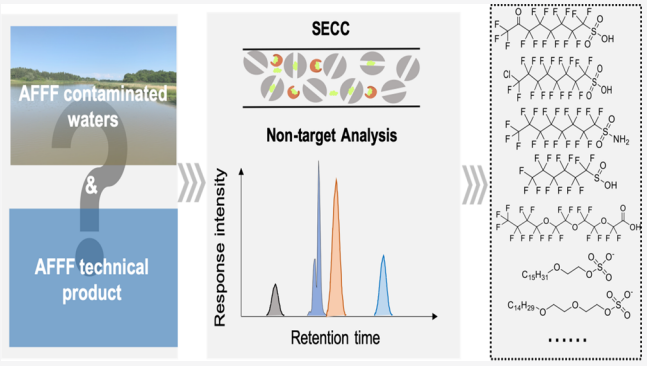
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ABSTRACT: More than 1000 per- and polyfluoroalkyl substances (PFASs) have been discovered by nontarget analysis (NTA), but their prioritization for health concerns is challenging. We developed a method by incorporating size-exclusion column co-elution (SECC) and NTA, to screen PFASs binding to human liver fatty acid binding protein (*h*L-FABP). Of 74 PFASs assessed, 20 were identified as *h*L-FABP ligands in which eight of them have high binding affinities. Increased PFAS binding affinities correlate with stronger responses in electrospray ionization (ESI^-) and longer retention times on a C18 column. This is well explained by a mechanistic model, which revealed that both polar and hydrophobic interactions are crucial for binding affinities. Encouraged by this, we then developed an SECC method to identify *h*L-FABP ligands, and all eight high-affinity ligands were selectively captured from 74 PFASs. The method was further applied to an aqueous film-forming foam (AFFF) product in which 31 new *h*L-FABP ligands were identified. Suspect and nontargeted screening revealed these ligands as analogues of perfluorosulfonic acids and homologues of alkyl ether sulfates (C_8- and C_{10}/EO_n, $C_8H_{17}(C_2H_4O)_nSO_4^-$, and $C_{10}H_{21}(C_2H_4O)_nSO_4^-$). The SECC method was then applied to AFFF-contaminated surface waters. In addition to perfluorooctanesulfonic acid and perfluorohexanesulfonic acid, eight other AFFF chemicals were discovered as novel ligands, including four C_{14}- and C_{15}/EO_n. This study implemented a high-throughput method to prioritize PFASs and revealed the existence of many previously unknown *h*L-FABP ligands.



INTRODUCTION

Per- and polyfluoroalkyl substances (PFASs) have been used in a variety of industrial and consumer applications including firefighting foams, insecticides, textile, and surface coatings in the past decades.[1−3] Of the PFASs studied to date, many are extremely stable and persistent in the environment due to their chemical structures with strong carbon−fluorine bonds. The broad distributions of PFASs in water, wildlife, air, and human blood have been reported worldwide.[3−6] Since the early 2000s, due to concerns regarding their persistence, bioaccumulation, and toxicity, perfluorooctanesulfonic acid (PFOS), perfluorooctanoic acid (PFOA), and related PFASs have been phased out in some parts of the world.[7] While these well-known classes of PFASs (e.g., perfluorinated acids) have been intensively studied and regulated by regulatory policies, the Toxic Substances Control Act (TSCA) Inventory lists 602 PFASs that are still actively used in U.S. commerce.[8] Identification of the unknown PFASs in the environment and their potential health risks is thus an urgent need.

Recent advances in high-resolution mass spectrometry-based nontarget analysis (NTA) provide an opportunity to characterize unknown PFASs in the environment.[9−12] The existence of >1000 PFASs in commercial products (e.g., aqueous film-forming foams, AFFFs)[12−14] and environmental matrices including surface waters,[11,15,16] wastewaters,[9] particulate matter,[10] and sediments[17] has been revealed by recent NTA studies. The detection of diverse PFASs in fish,[18] polar bears,[19] and human serum[20] provides evidence for the bioaccumulation of these PFASs from the environment to organisms. The bioaccumulation of previously unknown PFASs was also observed in laboratory-cultured rainbow trout exposed to AFFF technical products.[21] While these studies raised concerns about previously unknown PFASs, prioritization of these PFASs according to their potential health risks related to their bioaccumulation and toxicities is challenging due to their high number and diverse structures.

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Liver fatty acid binding protein (L-FABP) is a critical protein regulating the toxicokinetics of PFASs.[22−26] Previous studies have documented that the different binding affinities of classic PFASs to L-FABP contribute to their tissue distributions,[27] bioconcentration potential,[28] and placental transfers.[26] Physiologically based pharmacokinetic (PBPK) modeling also supports that L-FABP plays an essential role in determining the bioaccumulation of PFASs in organisms.[29,30] L-FABP is also a key transporter to delivering lipophilic ligands to peroxisome proliferator activated receptors (PPARs),[31,32] the nuclear receptors mediating certain toxicities of PFASs.[33−36] L-FABP may impact the toxicities of PFASs by directly regulating the signaling pathways of PPARs. For instance, sex-dependent impacts of perfluorononanoic acid (PFNA) on triglycerides in zebrafish liver were found to be mediated by L-FABP, via opposite regulations of expressions of L-FABP.[37] Regarding the critical role of L-FABP in toxicity and bioaccumulation, recent studies proposed to prioritize PFASs according to L-FABP binding affinities.[27,38] To address the challenge of prioritizing a large number of PFASs, which is infeasible for traditional bioassay-based toxicity tests, time- and cost-effective computational methods such as molecular dynamics simulations and molecular docking were proposed.[27,38−40] While in silico prediction is promising when a high-quality training set is available, the accuracy of prediction for unknown PFASs is challenged by their diverse structures. Thus, the implementation of a high-throughput experimental method to identify unknown PFASs binding to L-FABP is of great interest.

In this study, we aim to develop an experimental method for nontargeted screening of PFASs binding to *h*L-FABP from environmental mixtures including an AFFF technical product and AFFF-contaminated waters.[41] To achieve this, we implemented a method by employing size-exclusion chromatography-co-elution (SECC) and NTA to screen *h*L-FABP ligands with high selectivity; and the SECC method was then applied to an AFFF product and AFFF-contaminated river waters in which 39 AFFF chemicals were discovered as new *h*L-FABP ligands.

MATERIALS AND METHODS

Chemicals and Reagents. Seventy-four per- and polyfluoroalkyl substances (PFASs) (detailed information is provided in Table S1) were supplied by Evotec SE (Branford, CT) under contract to the U.S. Environmental Protection Agency (EPA) through a Material Transfer Agreement. The U.S. EPA's PFAS screening library was established with a chemical category-based prioritization approach through which the chemicals were selected based on structural diversity, toxicity data availability and quantity, physicochemical properties, and chemical standard availability,[42] providing a chance to evaluate structure-related binding affinities of PFASs to *h*L-FABP. A commercial AFFF technical product is representative AFFFs used in different air force bases in the United States, which was provided by the Davis−Monthan Air Force Base in Tucson (Arizona, USA) and manufactured by 3M as described in a previous study.[43] AFFF-contaminated water samples were collected from the west end of Lake Niapenco, Hamilton, Canada (details of sample collection are provided in the Supporting Information).[41] Water samples were collected using polypropylene tubes, which were then stored on ice during the transportation and extracted within 10 h. Other reagents, solvents, and protocols used are listed in the Supporting Information.

Binding Affinities of 74 PFASs to *h*L-FABP. The determination of the binding affinities of PFASs to *h*L-FABP was achieved by using a fluorescence displacement method. His-tagged *h*L-FABP was overexpressed in the BL 21 (DE3) strain of *E. coli*. The binding of 74 PFASs to *h*L-FABP was determined by the displacement of bound 1-anilinonaphthalene-8-sulfonic acid (1,8-ANS) from *h*L-FABP as previously described.[44] 1,8-ANS is nonfluorescent in aqueous buffer but becomes strongly fluorescent once it binds in a hydrophobic environment (i.e., proteins). Thus, the binding affinity of 74 PFASs to *h*L-FABP was measured by spiking each PFAS to the mixture of *h*L-FABP and 1,8-ANS. The decrease in fluorescence was expected as PFASs replaced 1,8-ANS from the hydrophobic protein binding pocket. More details about protein expression and binding affinity measurement are provided in the Supporting Information.

SEC-Co-elution (SECC) Method. The SEC-co-elution method was performed on an Agilent 1260 Infinity II HPLC (Agilent Technologies) with an AdvanceBio SEC column (300 Å/2.7 μm, Agilent Technologies). Solutions of 0.5 μM PFOS, PFOA, and 3:1 FTOH were spiked with 1 μM *h*L-FABP (n = 2) and incubated on ice for 60 min. Aliquots (100 μL) were loaded onto the SEC column. A 10 mM phosphate-buffered saline (PBS) solution was used as the mobile phase in the HPLC at a flow rate of 0.2 mL/min. The dual-wavelength signals were set to 260 and 280 nm to monitor proteins. Five fractions were collected using a fraction collector (Agilent 1260 Infinity II, Agilent Technologies) at a time interval of 2 min from 12 to 22 min. For chemical analysis, 1.2 mL of methanol was added to the fractions to denature proteins and extract PFASs. After centrifugation at 14,000g for 10 min, the supernatants were collected for PFAS analysis. The pellets of the fractions after centrifugation were subjected to sodium dodecyl sulfate−polyacrylamide gel electrophoresis (SDS-PAGE) or proteomics analysis to determine *h*L-FABP. For all subsequent experiments, *h*L-FABP was consistently eluted in fraction 2 (F2) (14−16 min).

To apply the SECC method to identify *h*L-FABP ligands from 74 PFASs, a 3M AFFF technical product, and AFFF-contaminated surface waters, the mixtures of chemicals were individually incubated with non-*h*L-FABP- and *h*L-FABP-overexpressing *E. coli* lysates. Specifically, 0.5 μM solutions of the 74 PFAS mixtures, a 25 mg L^{-1} solution of the AFFF technical product, and 1% SPE extracts of AFFF-contaminated water in buffer A (50 mM Tris-HCl, 150 mM NaCl, 1 mM dithiothreitol, pH 8.0) were used and incubated (n = 2) individually. The F2 fractions containing *h*L-FABP from SEC were extracted with methanol as mentioned above. The chemicals co-eluted with *h*L-FABP in F2 underwent mass spectrometry using an LC-Orbitrap for NTA to determine *h*L-FABP ligands. Details of the analysis of proteomes and small molecules including NTA analysis are provided in the Supporting Information.

Data Analysis. Statistical analyses were performed using GraphPad Prism (v7.0.4, GraphPad Software Inc., San Diego, CA, USA) or the R studio (v1.1.456, RStudio, Inc., Boston, MA, USA). MaxQuant algorithms (http://maxquant.org/, version 1.6.2.3) were used to process the proteomic raw files. The FASTA file of the human and *E. coli* proteome was downloaded from the UniProt database. The one-way ANOVA

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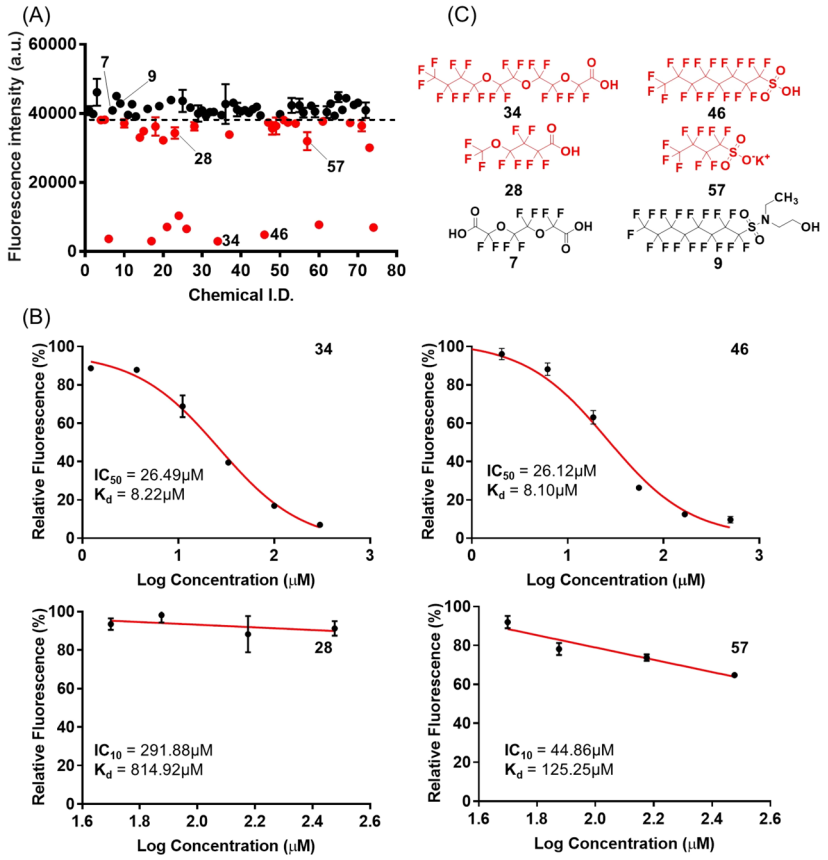


Figure 1. Characterization of the binding affinities of 74 PFASs to *h*L-FABP. (A) Fluorescence intensity of 1,8-ANS (40 μM) in 1 μM *h*L-FABP in the presence of individual PFASs at 300 μM. Inactive and active PFASs are labeled in black and red colors, respectively. (B) Dose-dependent competition of fluorescence signals of 1,8-ANS (40 μM) in 1 μM *h*L-FABP by four representative PFASs (**34**, **46**, **28**, & **57**). Fluorescence intensity was normalized to negative controls (1,8-ANS only). (C) Structures of four representative active PFASs (**34**, **46**, **28**, & **57**, red), and two inactive PFASs (**7** & **9**, black). PFAS **46**, **57**, and **9** refer to PFOS, PFHxS, and NEt-FOSE as described in Table S1.

test was used to detect the statistical significance of peak intensities or retention times between PFASs.

RESULTS AND DISCUSSION

Interactions between PFASs and *h*L-FABP Are Predictable by LC−MS Behaviors. To understand the structure-related interactions between PFASs and *h*L-FABP, the binding affinities of 74 structurally diverse PFASs from the U.S. EPA's PFAS screening library[42] to *h*L-FABP were tested through a fluorescence displacement method in which 1,8-ANS[23,24] was used as a fluorescence probe. The initial screening concentration of PFASs was fixed at 300 μM. While 44 of the 74 (59.5%) PFASs showed no fluorescence displacement (defined as "inactive"), 9 PFASs (12.2%) resulted in >50% reduction in fluorescence intensity (defined as "high" affinity ligands), and 21 (28.4%) PFASs, which induced less (10−50%) fluorescence intensity reduction, showed "weak" binding affinities (Figure 1A). The 30 positive hits were selected for the next round of quantification of their dissociation constant (K_d) by obtaining dose−response curves. Consistent with initial screening, low micrometer K_d values (4.3−18.7 μM) were detected for the nine high-affinity PFASs (Figure 1B). Among these PFASs, seven are classic perfluoroalkyl acids (PFAAs) and their salts including PFOS (**46**), PFOA, and PFNA (Figure 1C, Table S2), which have been well documented to bind strongly to L-FABP.[22,24,25] An ether chemical (**34**) was also detected with a high binding affinity (K_d = 8.2 μM). This result is supported by a very recent study, which determined that two other ether PFASs, 6:2 chlorinated polyfluorinated ether sulfonate (6:2 Cl-PFESA) and hexafluoropropylene oxide trimer acid (HFPO-TA), also have strong binding affinities to *h*L-FABP.[45]

In parallel, we analyzed the same 74 PFASs with an LC-Orbitrap in electrospray ionization negative mode (ESI^-), the most common ionization mode to analyze PFASs,[6,46,47] and found that 39 of them could not be detected due to the lack of ionizable functional groups (representative structures are shown in Figure 2A). Comparing with the results of the fluorescence displacement assay, all of these 39 "MS-inactive" PFASs also showed nonbinding to *h*L-FABP. In contrast with these MS-inactive PFASs, strong instrumental signals (indicated by high peak intensities) were observed for 20 *h*L-FABP ligands, especially for high-affinity ligands ($K_d \approx$ 4.3−18.7 μM). The average peak intensities of high-affinity PFASs (n = 9) are 3.7×10^8 at 1 μM, significantly ($p = 2.9 \times 10^{-21}$) higher than those of low-affinity (intensity = 8.2×10^7, n = 11) and inactive (intensity = 1.9×10^5, n = 54) PFASs (Figure 2B). A significant relationship was also observed between *h*L-FABP binding affinities of PFASs and their retention times on a C18 column. The retention times of high-affinity PFASs (4.47 ± 0.30 min, n = 9) are significantly ($p = 8.48 \times 10^{-5}$) longer than those of weak (3.26 ± 0.95 min, n = 11) and inactive PFASs (2.75 ± 0.65 min, n = 15) (Figure 2C).

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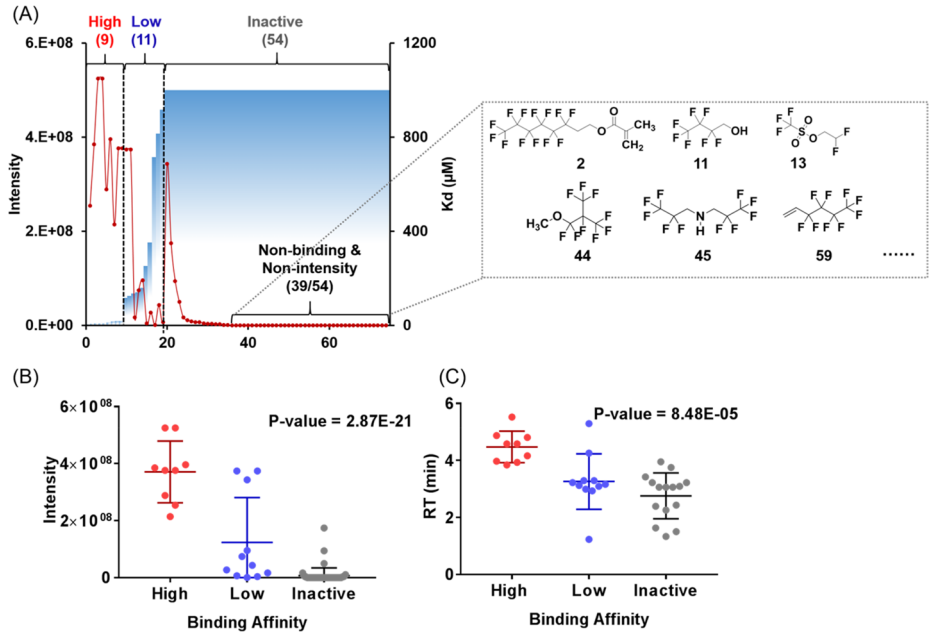


Figure 2. Binding affinities of 74 PFASs to *h*L-FABP are predictable by their LC−MS behaviors. (A) Peak intensities (red dots) and K_d (blue columns) values of 9 high-affinity ($K_d \approx 4.3-18\ \mu M$), 11 low-affinity ($K_d \approx 116-916\ \mu M$), and 54 inactive ($K_d > 1000\ \mu M$) PFASs. Note that all 39 MS-inactive PFASs do not bind to *h*L-FABP. Structures of representative MS-inactive PFASs are shown in the box. (B) Peak intensities of high-affinity PFASs are significantly higher than those of weak and inactive PFASs. *p* values were determined by one-way ANOVA. (C) Retention times of high-affinity PFASs on C18 columns are significantly longer than those of weak and inactive PFASs. Thirty-nine PFASs, which were not detected by LC−MS, are not shown in (C). *p* values were determined by one-way ANOVA.

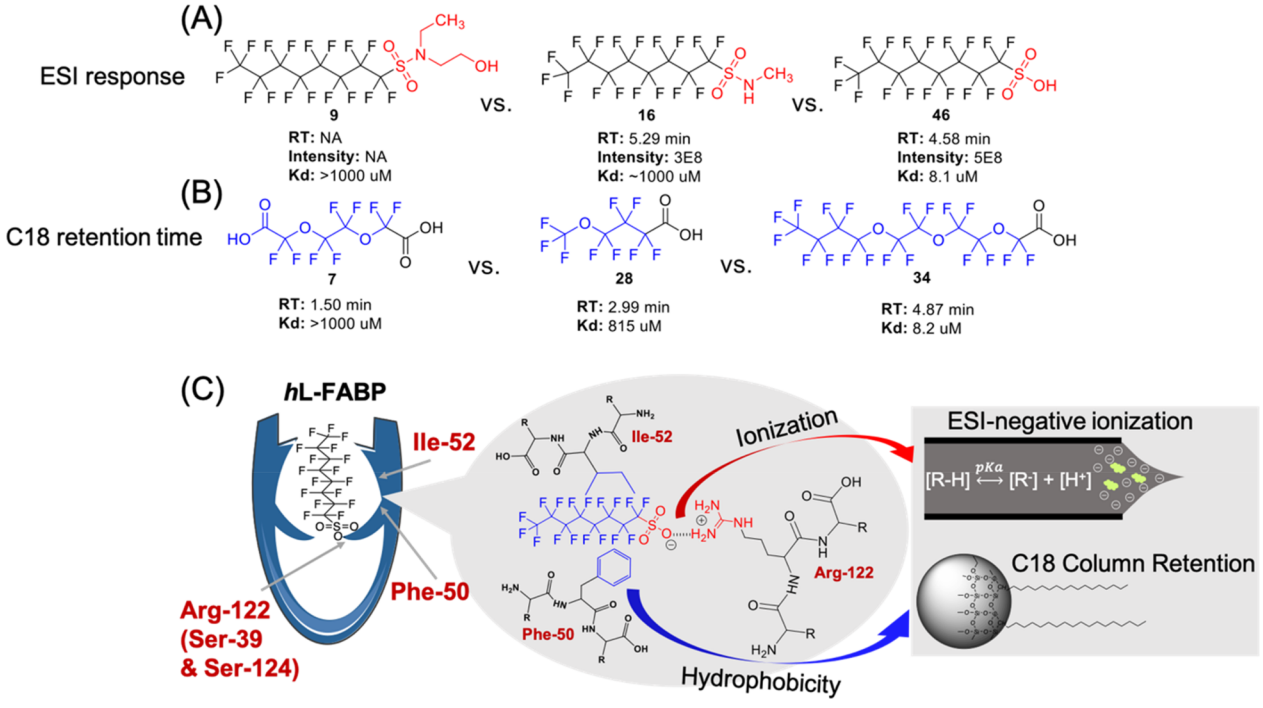


Figure 3. Proposed mechanisms to link LC−MS behaviors and *h*L-FABP binding affinities. (A) Binding affinities of three PFOS analogues are related to their ESI^- responses. (B) Binding affinities of three ether analogues are related to their retention times on C18 columns. (C) Proposed physicochemical mechanisms to link LC−MS behaviors and *h*L-FABP binding affinities. The polar interactions (red color) between the negatively charged acidic group and positively charged Arg-122 on *h*L-FABP mimic the process of ionization of PFASs under ESI^-; the hydrophobic interactions (blue colors) between alkyl chains and Phe-50 and Ile-52 mimic the partition of PFASs on the C18 column. PFAS **9**, **16**, and **46** refer to NEt-FOSE, NMe-FOSA, and PFOS as described in Table S1.

The association of *h*L-FABP binding affinities with LC−MS behaviors of PFASs is clearly illustrated by the representative structurally similar analogues in Figure 3. The instrumental responses were increased in the order NEt-FOSE (**9**) < NMe-FOSA (**16**) < PFOS (**46**) (Figure 3A), mainly due to the decrease in their pK_a values. Such a trend is consistent with the

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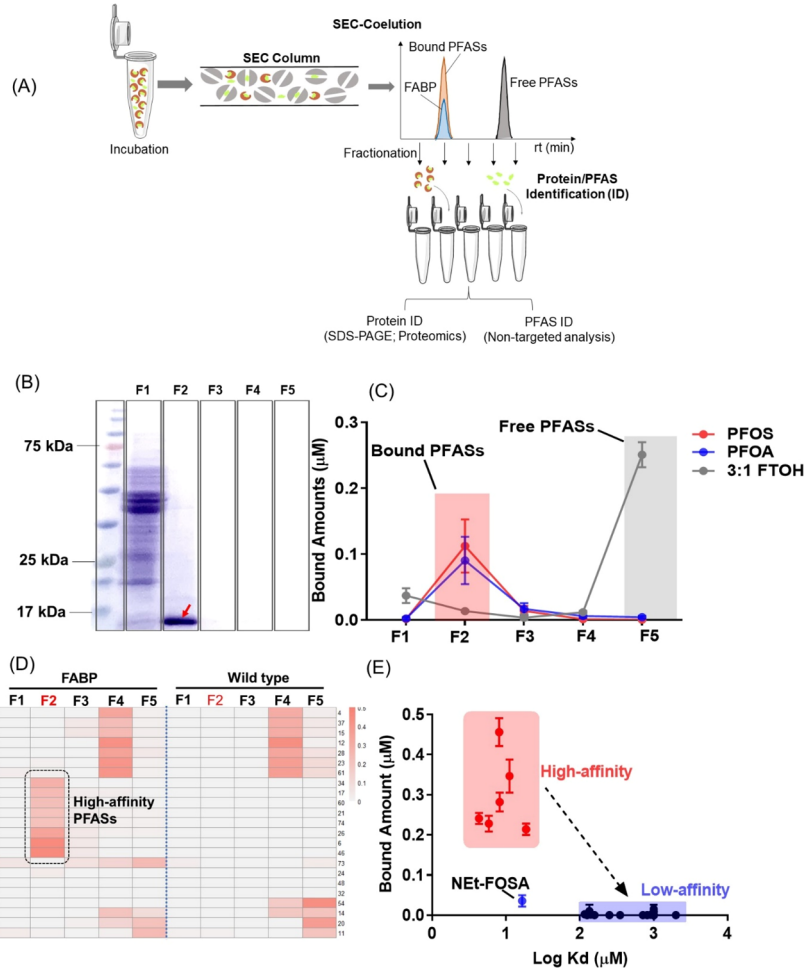


Figure 4. Benchmarking the SECC method to identify PFASs binding to *h*L-FABP. (A) Workflow of the SECC method. *h*L-FABP ligands (bound) and inactive PFASs (free) were separated by SEC according to their sizes. Proteins and PFASs in each HPLC fraction were analyzed by proteomics and nontarget analysis to assign the interactions. (B) *h*L-FABP was eluted in F2 (14−16 min) from SEC as evidenced by SDS-PAGE. Red color arrow indicates the band of *h*L-FABP (~14 kDa). (C) Co-elution of PFOA and PFOS with *h*L-FABP in F2. Note that 3:1 FTOH (inactive) was eluted in F5 (20−22 min) as a free chemical. (D) Benchmarking the SECC method with a mixture of 74 PFASs (0.5 μM for each chemical). Note that only eight high-affinity PFASs were co-eluted with *h*L-FABP in F2, compared to non-*h*L-FABP lysates. (E) SECC method selectively captured the high-affinity PFASs. NEt-FOSA (blue dot) was not captured by the SECC method even though it was identified as a high-affinity *h*L-FABP ligand by the fluorescence displacement method. Only six individual high-affinity PFASs are shown (two of the *h*L-FABP ligands are salts of PFOA and PFOS).

stronger binding affinities of PFOS (K_d = 8.1 μM) relative to NMe-FOSA (K_d ≈ 1000 μM) and NEt-FOSE (K_d > 1000 μM). Similarly, the increasing retention times of three analogues of ether chemicals **7**, **28**, and **34** are in accordance with their increasing binding affinities to *h*L-FABP. Moreover, the crystal structure of *h*L-FABP complexed with palmitic acid shows that *h*L-FABP is capable of binding palmitic acids at two sites.[48] The ionic and hydrogen bonds between the carboxylic acid group of palmitic acid and Arg-122, Ser-39, and Ser-124, together with the hydrophobic interactions between the alkyl chain and interior cavity, were revealed to play vital roles in stabilizing the complex at both binding sites. Although the crystal structure of PFASs binding to *h*L-FABP is not resolved, molecular docking simulations revealed that both polar and hydrophobic interactions play important roles.[22,25,38]

Therefore, together with the experimental and in silico information, we proposed a mechanistic model to link LC−MS behaviors to *h*L-FABP binding affinities of PFASs (Figure 3C): the polar interaction between PFASs and *h*L-FABP relies on the negative charge of acidic head groups to maintain ionic and hydrogen bonds with Arg-122, Ser-39, and Ser-124,[22] and the responses of PFASs in an ESI$^-$ source are related to their pK_a-dependent ionization efficiencies; moreover, the binding of the alkyl chain of PFAS to the hydrophobic cavity of *h*L-FABP (i.e., Phe-50, Ile-52, etc.) and the retention of PFASs on the C18 stationary phase are both determined by their hydrophobicity. To the best of our knowledge, this is the first study to connect LC−MS behaviors to bioactivities of PFASs. The discovery provides a straightforward strategy to prioritize the PFASs detected by NTA: only PFASs detected under ESI$^-$ and eluted after 3.0 min (in current LC conditions referred to in the Supporting Information) are possible *h*L-FABP ligands.

Note that this is a descriptive model to initially narrow down priority PFASs. However, not all PFASs meeting the LC−MS criteria (RT > 3.0 min under ESI$^-$) are necessarily *h*L-FABP ligands. For instance, NMe-FOSA (chemical **16** in Figure 3A) was detected under ESI$^-$ with high intensity (3.43×10^8 at 0.5 μM); however, it is a weak *h*L-FABP ligand with K_d ≈ 1000 μM. Since the estimated pK_a value of NMe-FOSA is 8.78,[49] it is neutral at physiological pH (7.4) and is not expected to form

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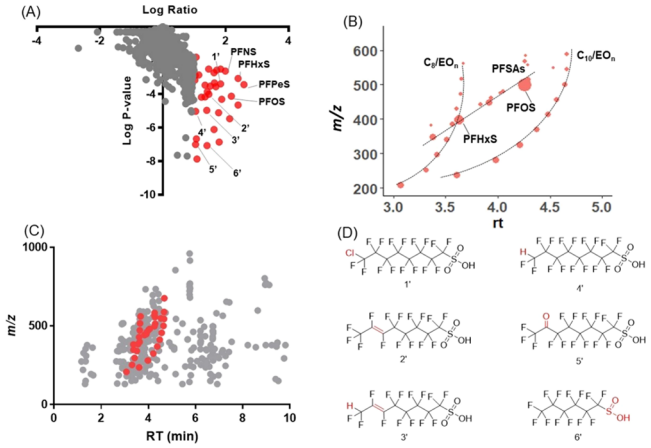


Figure 5. Nontarget screening of PFASs binding to *h*L-FABP in the AFFF technical product. (A) Volcano plot representing the log-transformed ratios and corresponding *p* values of features detected under ESI$^-$. The ratio of peak abundances of PFASs detected in *h*L-FABP-overexpressing *E. coli* lysates to those in non-*h*L-FABP lysates was calculated as described in the Method section in the Supporting Information. The red dot indicates features significantly co-eluted with *h*L-FABP in F2. (B) Retention times and *m/z* values of *h*L-FABP ligands detected under ESI$^-$. Note that all ligands are grouped into three classes. (C) Comparison of the *m/z* and retention times of *h*L-FABP ligands (red color) to all AFFF chemicals (gray color) detected under ESI$^-$. (D) Structures of six novel *h*L-FABP ligands as PFOS analogues, at confidence level 3. Note that multiple isomers may exist but only one representative structure from the PFAS library[53] was shown for each chemical.

a polar interaction with Arg-122, Ser-39, and Ser-124 on *h*L-FABP. Moreover, the instrumental responses of nitrogen-containing chemicals were reported to be insensitive to pK_a, and all triazines with pK_a < 10 are efficiently ionized under ESI$^-$.[50] In addition, previous studies have reported the decrease in *h*L-FABP binding affinities for longer-chain PFCAs when their carbon chain lengths are greater than 12.[22] Therefore, the model may not be applied to the PFASs with extremely long carbon chains since they cannot fit into the binding pocket of *h*L-FABP. This highlights the need to develop a selective method to further identify *h*L-FABP ligands that were initially narrowed down by the LC−MS criteria.

Development of an SEC-Co-elution Method to Screen Unknown *h*L-FABP Ligands. We opted to develop an SEC co-elution (SECC) method to identify *h*L-FABP ligands by incorporating SEC fractionation and nontarget analysis in ESI negative mode (Figure 4A). In brief, the *h*L-FABP proteins were incubated with PFASs and then loaded onto an SEC column for nondenaturing fractionations. The high-affinity PFASs binding to *h*L-FABP (eluted earlier) were separated from free unbound inactive PFASs (eluted later) according to their sizes. The fractions from the SEC were collected and analyzed by proteomics and nontarget analysis to determine the ligands co-eluted with *h*L-FABP. To reduce the impacts of nonspecific binding, we opted to use crude *E. coli* lysates rather than purified *h*L-FABP for the SECC, as nonspecific bound PFASs will be removed by higher-abundance *E. coli* proteins.

For the proof of concept, we used PFOS (K_d = 8.1 μM), PFOA (K_d = 18.8 μM), and 3:1 FTOH (K_d > 1000 μM) with different binding affinities, which were determined by the fluorescence displacement experiment, to verify the SECC method. Solutions (0.5 μM) of the three PFASs were individually incubated with 1 μM *E. coli* lysates overexpressing *h*L-FABP or non-*h*L-FABP-overexpressing *E. coli* lysates and then fractionated with an SEC column. SDS-PAGE revealed that *h*L-FABP (~14 kDa) was eluted in fraction 2 (F2, 14−16 min), while other background *E. coli* proteins were eluted in earlier fractions (F1, 12−14 min) due to their bigger sizes (Figure 4B). The presence of *h*L-FABP in F2 was confirmed with proteomics (Figure S5). One of the two *h*L-FABP ligands, PFOS, was co-eluted with *h*L-FABP in F2 at 0.11 ± 0.028 μM (Figure 4C). The other ligand, PFOA, was also co-eluted in F2, albeit at a smaller amount (0.09 ± 0.025 μM), partially due to its weaker affinity relative to PFOS. In contrast, the inactive 3:1 FTOH was eluted later in F5 as a free chemical. For the non-*h*L-FABP lysates, all three PFASs were eluted in F5 as free chemicals. These results verified the selectivity of the SECC method to identify *h*L-FABP ligands.

To further benchmark the method to identify *h*L-FABP ligands from a mixture, we mixed all 74 PFASs at 0.5 μM and then subjected the mixture to the ligand identification. The majority of weak or inactive PFASs were eluted in F4 or F5 as free chemicals, which were observed in both *h*L-FABP-overexpressing and non-*h*L-FABP lysates (Figure 4D). Eight PFASs were eluted in F2 when incubating with *h*L-FABP-overexpressing lysates but not in non-*h*L-FABP lysates, clearly suggesting the binding of these PFASs to *h*L-FABP. This result was consistent with the fluorescence displacement experiment through which all eight of these PFASs were defined as strong *h*L-FABP ligands, such as PFOA (**74**), PFNA (**17**), and FOSA (**60**) (Table S1). The bound amounts of these high-affinity PFASs co-eluted in F2 were 0.21−0.46 μM, while they were <0.01 μM for other low-affinity PFASs (Figure 4E). Interestingly, NEt-FOSA, which was also determined as a strong *h*L-FABP ligand by the fluorescence displacement method, was co-eluted in F2 with a minor amount (blue dots in Figure 4E). To verify the inconsistency between the two methods, we used LC−MS to analyze the standard of NEt-FOSA, and PFOS was detected as a high-percentage (30%) impurity, which mainly explained the high binding affinity of NEt-FOSA to *h*L-FABP in the fluorescence displacement experiment (Figure S6). Therefore, the results indicated that NEt-FOSA is not a strong *h*L-FABP ligand and its exclusion from the screening assay was caused by the PFOS impurity in the sample. Together with the results, we conclude that SECC is a highly selective method to identify *h*L-FABP ligands from a mixture. Effect-directed analysis (EDA) is the predominant strategy to identify causative chemicals from the environmental mixture.[51,52] However, EDA is prone to false-positive rates due to the co-elution of hundreds of chemicals in the same fraction. In contrast, the SECC method is very effective in identifying unknown ligands even if multiple ligands are present at the same time. We proposed that SECC might be a supplementary method to EDA for the identification of unknown ligands in the environment, and the EDA method could be further employed to quantify the contributions of individual ligands.

Sulfonic Acid Analogues Were Identified as the Major *h*L-FABP Ligands in an AFFF Product. We then applied the SECC method to identify PFASs binding to *h*L-FABP in a 3M AFFF technical product. A 25 mg L^{-1} (~50 μM) solution of an AFFF product was incubated with 1 μM *E. coli* lysates overexpressing *h*L-FABP or non-*h*L-FABP lysates for the SECC method. Nontarget analysis was employed to determine the chemicals eluted in F2 from *h*L-FABP-

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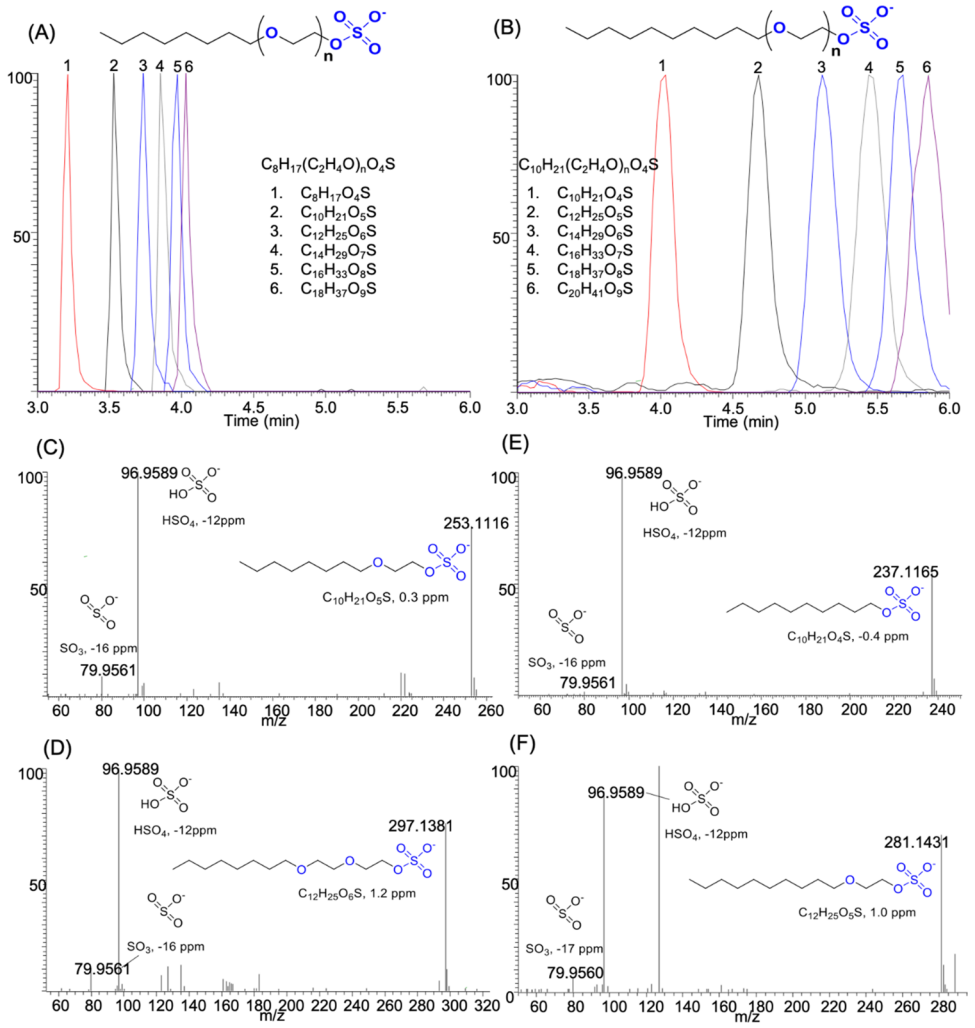


Figure 6. Alkyl ether sulfates were identified as predominant classes of *h*L-FABP ligands in the AFFF technical product. (A) Chromatograms of six identified C_8/EO_n and their assigned formulas. (B) Chromatograms of six identified C_{10}/EO_n and their assigned formulas. (C) MS^2 spectra and assigned fragments of C_8/EO_n **2**. (D) MS^2 spectra and assigned fragments of C_8/EO_n **3**. (E) MS^2 spectra and assigned fragments of C_{10}/EO_n **1**. (F) MS^2 spectra and assigned fragments of C_8/EO_n **2**.

overexpressing and non-*h*L-FABP lysates. Only chemicals showing significantly higher abundances (10-fold, $p < 0.05$) in *h*L-FABP-overexpressing lysates than those in non-*h*L-FABP lysates were considered as *h*L-FABP ligands. Among 4056 features detected under ESI^-, 33 chemicals were specifically co-eluted with *h*L-FABP in F2 (Figure 5A) after excluding isotopic peaks and adducts. Three major classes of homologues were clearly observed according to their stepwise increase in *m/z* and retention times (Figure 5B). Corresponding to the LC−MS mechanistic model proposed above, the retention times of these detected *h*L-FABP ligands fall into a narrow range of 3.07−4.66 min, while a wide range of retention times was detected for 544 other AFFF chemical features (Figure 5C). This confirms the above results that only PFASs with longer retention times (>3 min) exhibit sufficient binding affinities to *h*L-FABP; at the same time, large PFASs with extremely long retention times (>5 min) may also exhibit decreased binding affinities due to the limited size of the *h*L-FABP binding pocket.

To get exact chemical identities for these detected *h*L-FABP ligands, we searched against the PFAS library established in Liu's study,[53] which summarized 1021 PFASs in more than 130 classes discovered in previous NTA studies (details of suspect screening are provided in the Supporting Information). Eleven PFASs were identified with this strategy, and all of them are analogues to perfluorosulfonic acids (PFSAs). PFOS and perfluorohexanesulfonate (PFHxS), two high-affinity *h*L-FABP ligands, were identified as the two most abundant ligands according to their peak intensities (Figure 5B). Several other classic odd-chained PFSAs, including perfluoropentanesulfonate (PFPeS), perfluoroheptanesulfonate (PFHpS), and perfluorononanesulfonate (PFNS), were detected with lower abundances, and the identities of PFPeS and PFHxS were confirmed by authentic standards (Figure S7). In addition to these legacy PFSAs, several other PFOS analogues (Figure 5D) including chlorinated (**1′**), unsaturated (**2′**), or hydrogenated PFOS (**3′**) were also reproducibly detected across replicates. Except for PFHxS and PFOS, 9 of the 11 PFSAs detected as *h*L-FABP ligands by the SECC method have never been reported before. Due to the lack of authentic standards, seven of the detected ligands were assigned at confidence level 2 or 3 according to the scale by Schymanski et al.,[54] by exact mass, MS^2 spectra, and isotopic peaks. Further studies are warranted to synthesize standards to determine their binding affinities.

Although PFSAs were identified, two other classes of homologues (Figure 5B) were not identified in the PFAS

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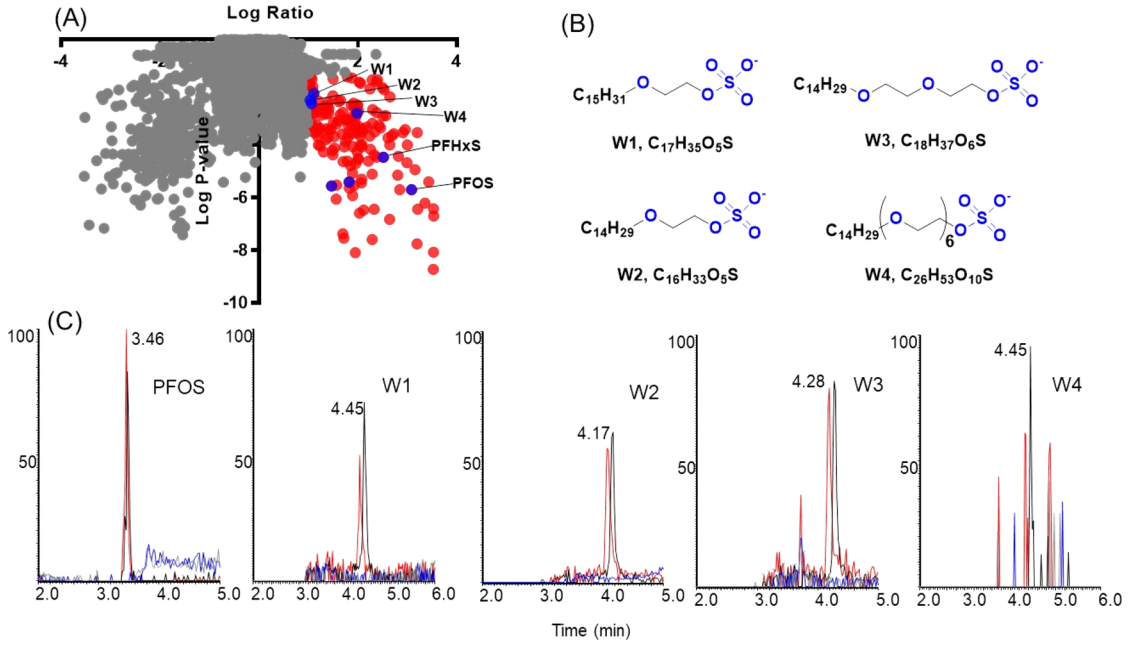


Figure 7. Nontarget screening of *h*L-FABP ligands in AFFF-contaminated river waters. (A) Volcano plot representing the log-transformed ratios and corresponding *p* values of features detected. The red dot indicates features significantly co-eluted with *h*L-FABP in F2. The blue dots indicate PFASs or AFFF surfactants detected. (B) Structures of four detected C_{15}- and C_{14}-ether sulfates. (C) Chromatograms of PFOS and four alkyl ether sulfates. Red and black colors indicate the chromatograms from two replicates of *h*L-FABP-overexpressing lysates. Blue and gray colors indicate the chromatograms from two replicates of non-*h*L-FABP lysates.

library. We hypothesized that these chemicals might be nonfluorinated AFFF chemicals, which comprised the major portion of AFFF technical products (PFASs only comprise 0.9−1.5% w/w).[55] A very recent suspect screening study identified nine classes of hydrocarbon surfactants in AFFFs.[55] Thus, we searched against the AFFF surfactant library (869 compounds)[55] and found two of the *h*L-FABP ligands, identified as $C_8H_{17}O_4S$ and $C_{10}H_{21}O_4S$ alkyl sulfates. Interestingly, these two alkyl sulfates belonged to two unidentified classes of *h*L-FABP ligands. The chromatograms of the homologues of $C_8H_{17}O_4S$ were obtained (Figure 6A), and their formulas were predicted as $C_8H_{17}(C_2H_4O)_nO_4S$, the alkyl ether sulfates (C_8/EO_n). The prediction of structures was also supported by their high-resolution MS^2 spectra in which HSO_4^- and SO_3^- fragments were observed as characteristic fragments of sulfate (Figures 6C,D). Homologues of $C_{10}H_{21}O_4S$ were detected at longer retention times than C_8/EO_n (Figure 6B) due to their longer hydrophobic alkyl chains. MS^2 spectra (Figures 6E,F) further confirmed $C_{10}H_{21}O_4S$ homologues as C_{10}/EO_n. The identification of C_8- and C_{10}/EO_n as the predominant *h*L-FABP ligands was consistent with the structure-related activities of the 74 PFASs we established in the fluorescence displacement experiment of which the ether analogue (**34**) of perfluorinated carboxylic acids was demonstrated to be a strong *h*L-FABP ligand.

Predominance of Sulfonic Acid Analogues in AFFF-Contaminated River Waters Is Evidenced by the SECC Method. We then applied the SECC method to identify PFASs binding to *h*L-FABP in AFFF-contaminated surface waters, collected from the Welland River next to the Hamilton International Airport.[41] Consistent with previous studies,[41,56] high concentrations of PFOS (232 ng/L) were detected in water samples collected from the Welland River. The extracts from river waters from C_{18} SPE cartridges were spiked with 1 μM *E. coli* lysates overexpressing *h*L-FABP and non-*h*L-FABP lysates for the SECC method. Two hundred and seven chemicals were detected as putative ligands of *h*L-FABP as shown in Figure 7A. Only 10 of them were assigned by searching against AFFF, PFAS, and surfactant libraries, including two PFASs (PFOS and PFHxS) and eight alkyl ether sulfates. Since it has been reported that *h*L-FABP may bind to many other endogenous metabolites,[57] this may explain the majority of *h*L-FABP ligands we detected in the natural waters. The detection of PFOS and PFHxS as the only fluorinated ligands corresponded to the results of the AFFF technical product in which PFOS and PFHxS are the two most abundant fluorinated ligands. Other PFASs, such as PFOA, were not detected in the contaminated water, which may be due to their 10−100 times lower concentrations than PFOS.[41,56]

Interestingly, the alkyl ether sulfates were also detected by the SECC method as the ligands of *h*L-FABP in AFFF-contaminated waters. This was consistent with the results of the AFFF technical product. However, C_8- and C_{10}/EO_n were detected as the only ligands in the AFFF product, whereas C_{14}- and C_{15}/EO_n were detected as the major ligands in river waters (Figures 7,C). This could be due to the components of the AFFF used at the Hamilton Airport differing from the 3 M AFFF technical product we used in the current study. Differences in the components of AFFF products have been well documented in previous studies.[12,21,55] Alternatively, alkyl ether sulfates are increasingly used in North America as surfactants.[58] The identification of different alkyl ether sulfates in the river water from the AFFF product may indicate AFFF-independent sources of these compounds. The identification of alkyl ether sulfates as *h*L-FABP ligands is significant since a large amount of these surfactants is present in the AFFF technical product, and they were also reported to occur in groundwater.[55]

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Implications. Current studies on nontarget analysis and toxicities of PFASs are disconnected due to the challenges posed by the large numbers (>1000) and diverse structures of PFASs. The SECC method provides a high-throughput experimental way to tackle the challenge of prioritizing PFASs according to key proteins, especially when their authentic standards are not available. While this study is focused on *h*L-FABP due to its critical role in regulating the toxicokinetics of PFASs, the protein-centric method could also be adapted to screen PFASs binding to other key proteins, such as PPARs. Computational toxicology is the predominant strategy for high-throughput predictions of toxicities of chemical contaminants. However, these models are prone to uncertainties, especially when the training dataset is inappropriate. The SECC method provides an opportunity to identify ligands even if prior information on binding schemes is limited (e.g., alkyl ether sulfates). We believe the SECC method would be valuable, in combination with computational toxicology, to uncover previously unknown PFASs binding to key proteins. However, one of the limitations of the current study is that only a single AFFF technical product and AFFF-contaminated waters were applied. Since previous studies have well documented the variation in chemical components across AFFF products,[12] the application of the SECC method to more AFFF-contaminated environmental samples is of ongoing interest.

The present study revealed the existence of many previously unknown, AFFF-derived *h*L-FABP ligands. Unexpectedly, nonfluorinated AFFF chemicals were identified as an even bigger class of *h*L-FABP ligands. Indeed, these nonfluorinated ionic surfactants were proposed as environmentally safe chemicals to replace PFASs or organic solvents, but limited data is available regarding their potential toxicities.[58−60] Here, we highlighted *h*L-FABP as a potential critical target for these nonfluorinated ionic surfactants.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.0c00049.

Additional information of the experimental procedures; western blot and SDS-PAGE results of His-tagged hL-FABP; direct binding of 1,8-ANS to hL-FABP; calculation process of dissociation constants (K_d) of PFASs via dose−response curves obtained using the fluorescence displacement assay; verification of hL-FABP in SEC fraction F2 by proteomics; PFOS impurity in NEt-FOSA standard measured by the fluorescence displacement method; circular dichroism (CD) spectra of the recombinant poly-His tagged hL-FABP 253 protein and their high-tension voltage; table of the summary of 74 per- and polyfluoroalkyl substances (PFASs); table of the inhibition concentration (IC50/10), dissociation constants (K_d), LC-MS peak intensities, and retention times of 19 PFASs binding to hL-FABP; and table of the exact masses, retention times, and predicted formulas and structures of 11 PFASs (PDF)

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Notes

The authors declare no competing financial interest.

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